YEAR 9 SCIENCE
CHEMICALS SCIENCES
TEST 2 - CHEMICAL REACTIONS

NAME: SOLUTIONS **CLASS:** ______ **Mark:** —/33

Chemical reactions, including combustion and the reactions of acids, are important in both non-living and living systems and involve energy transfer.

Multiple Choice **Write the answer to each question in the appropriate box at right.**

QUESTION	ANSWER
1	A
2	A
3	C
4	B
5	B
6	A
7	A
8	A
9	A
10	B
11	B
12	C

1. Which of the following word equations correctly explains an acid reacting with a metal?

   (a) zinc + hydrochloric acid → zinc chloride + hydrogen
   (b) magnesium + sulphuric acid → magnesium chloride + hydrogen
   (c) magnesium + hydrochloric acid → carbon dioxide + hydrogen
   (d) zinc + hydrochloric acid → zinc chloride + carbon dioxide

2. Which of the following word equations correctly explains an acid reacting with a carbonate?

   (a) hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide
   (b) hydrochloric acid + calcium carbonate → calcium chloride + water + hydrogen
   (c) sulphuric acid + calcium carbonate → calcium chloride + water + carbon dioxide
   (d) sulphuric acid + calcium carbonate → calcium sulphate + hydrogen + carbon dioxide

3. Which of the following is ***not*** a property of acids?

   (a) React with some metals to produce hydrogen gas.
   (b) Turn litmus red.
   (c) Sweet taste.
   (d) Sour taste.

4. Which of the following is a ***base***?

   (a) NaCl (sodium chloride)
   (b) NaOH (sodium hydroxide)
   (c) CH_3OH (methanol)
   (d) HCl (hydrochloric acid)

5. Which of the following could be the pH for a pool that had turned ***slightly acidic***?

   (a) 2
   (b) 6.5
   (c) 7.5
   (d) 7

6. Which of the following is the correct ***general equation*** to describe the reaction of an acid and a base?

   (a) acid + base → a salt + water
   (b) acid + base → metal hydroxide + carbon dioxide
   (c) acid + base → carbon dioxide + water
   (d) acid + base → hydrogen + water

7. Magnesium hydroxide can be taken to settle stomach acids. The reason why it is useful for this is because:

   (a) the magnesium hydroxide neutralises the stomach acid.
   (b) the magnesium hydroxide is not very acidic so it won't make your stomach more acidic.
   (c) magnesium hydroxide is a base so it will turn the stomach acid into a salt.
   (d) stomach acid will turn into a base when the magnesium hydroxide is taken.

8. Which of the following is most likely to be the pH of lemon juice?

   (a) 3
   (b) 7
   (c) 10
   (d) 14

9. Which of the following will turn universal indicator ***red***?

   (a) Vinegar.
   (b) Sodium hydroxide.
   (c) Distilled water.
   (d) Glucose solution.

10. A chemist tells you that a substance she has tested has a pH of 1. From this you know:

    (a) its colour.
    (b) that it is a strong acid.
    (c) that it is a weak base.
    (d) that it is concentrated.

11. Which of the following facts about toothpaste explain why it helps reduce tooth decay?

    (a) It has a minty taste.
    (b) It is slightly basic.
    (c) It contains fluoride.
    (d) It is abrasive.

12. The test for determining whether a colourless, odourless gas is hydrogen is called the:

    (a) glowing splint test.
    (b) pH test.
    (c) pop test.
    (d) limewater test.

Short Answer **Write the answer to the questions in the spaces provided.**

1. List ***two*** characteristics of acids and bases. (Any 2 for each - 1 mark each.)

Acids	**Bases**
• Produce H^{1+} ions in solution.	• Produce OH^{1-} ions in solution.
• Have a sour taste.	• Any metal oxide or hydroxide.
• Turn blue litmus red.	• Have a soapy feel.
	• Have a bitter taste.
	• Turn red litmus blue.

(4)

2. Classify the following compounds as **ACID, BASE** or **SALT**.

H_2SO_4 $Mg(OH)_2$ sodium chloride Al_2O_3 HNO_3 $CuSO_4$ calcium oxide HCl

ACID	BASE	SALT
H_2SO_4 HNO_3 HCl	$Mg(OH)_2$ Al_2O_3 Calcium oxide	sodium chloride $CuSO_4$

($\frac{1}{2}$ mark off each error.)

(4)

3. Complete the following table, giving ***estimates on the pH where necessary***, the ***colour of the universal indicator*** and whether the substance is a ***strong/weak acid, a strong/weak base or neutral***.

SUBSTANCE	pH	UNIVERSAL INDICATOR (colour)	ACID / BASE/ NEUTRAL (strong / weak)
vinegar	5	orange / yellow	weak acid
soda water	3	light red	weak acid
2.00 M H_2SO_4 (sulphuric acid)	0/1	red	strong acid
1.00 M Na_2SO_4 (sodium sulphate)	8	light blue	weak base
sugar solution	7	green	neutral
yoghurt	5/6	yellow	weak acid

(6)

accept either.

($\frac{1}{2}$ mark off each error.)

4. Complete the following general word equations for acid reactions.

ACID + METAL → SALT + HYDROGEN

ACID + CARBONATE → SALT + WATER ($\frac{1}{2}$ mark off each error.)

+ CARBON DIOXIDE

(3)

5. Complete the following **WORD EQUATIONS** for acid reactions, naming the salt produced and other products.

(a) magnesium hydroxide + sulphuric acid → magnesium sulphate (1) + water (1)

(b) zinc + nitric acid → zinc nitrate (1) + hydrogen (1)

(4)

6. ***Optional***

Balance the following equations.

(a) $CaCO_{3\,(s)}$ + 2 $HCl_{(aq)}$ → $CaCl_{2\,(aq)}$ + $H_2O_{(l)}$ + $CO_{2\,(g)}$ (2)

(b) $CH_{4\,(g)}$ + 2 $O_{2\,(g)}$ → $CO_{2\,(g)}$ + 2 $H_2O_{(g)}$ (2)

(4)